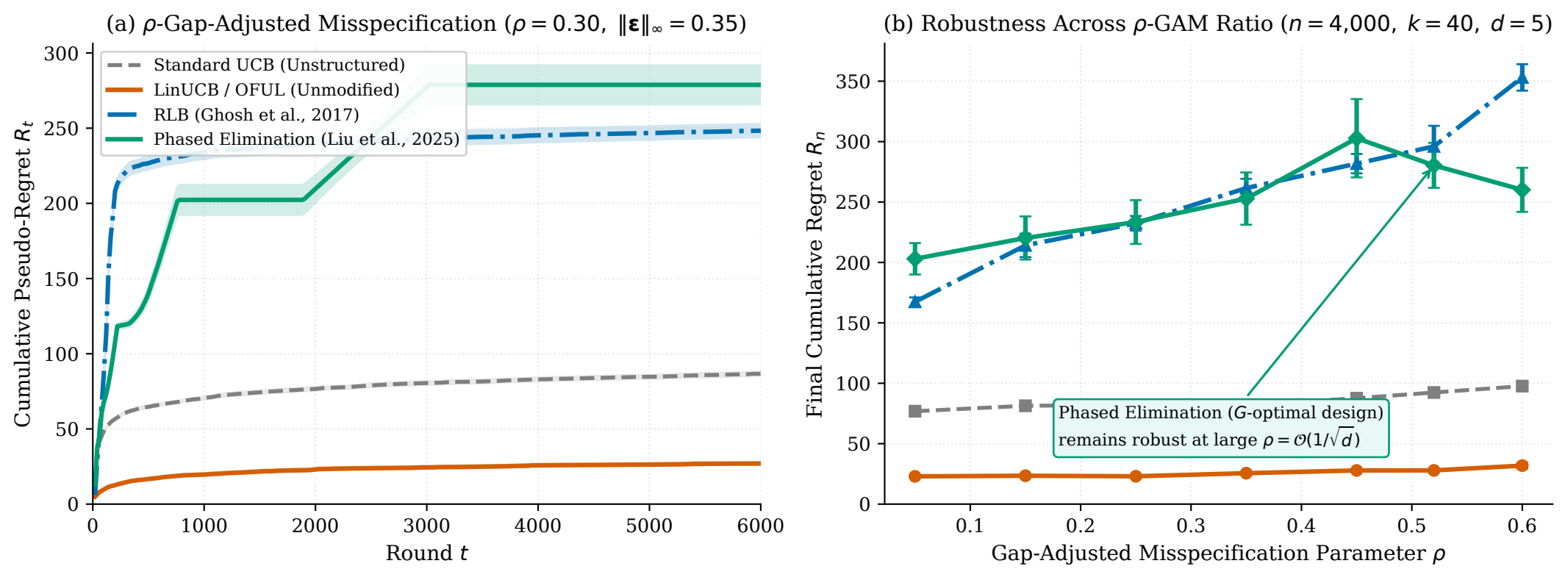

(a) ρ-Gap-Adjusted Misspecification ($\rho = 0.30$, $\|\boldsymbol{\varepsilon}\|_\infty = 0.35$)
Standard UCB (Unstructured)
LinUCB / OFUL (Unmodified)
RLB (Ghosh et al., 2017)
Phased Elimination (Liu et al., 2025)
Cumulative Pseudo-Regret R_t
Round t
0
50
100
150
200
250
300
0
1000
2000
3000
4000
5000
6000
(b) Robustness Across ρ-GAM Ratio ($n = 4{,}000$, $k = 40$, $d = 5$)
Final Cumulative Regret R_n
Gap-Adjusted Misspecification Parameter ρ
Phased Elimination (G-optimal design)
remains robust at large $\rho = \mathcal{O}(1/\sqrt{d})$
0
50
100
150
200
250
300
350
0.1
0.2
0.3
0.4
0.5
0.6